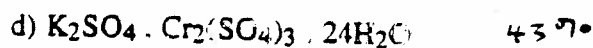
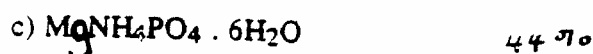
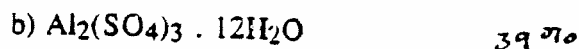


SCH - 3A CHEMISTRY

HYDRATE PROBLEMS

1. Determine the percentage by mass of water in each of the following hydrates:



2. Upon heating 2.191 g of the hydrate of calcium chloride, 1.110 g of residue were formed. What is the formula of the hydrate? $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$
3. 1.362 g of anhydrous calcium sulfate were produced from heating 1.722 g of the hydrate. What is the formula of the hydrate? $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
4. Vigorous heating of 2.465 g of the hydrate of magnesium sulfate yielded 1.204 g of residue. What is the formula of the hydrate? $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
5. 2.195 g of the hydrate of zinc acetate produces 1.835 g of residue. Determine the formula of the hydrate. $\text{Zn}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 2\text{H}_2\text{O}$